socket on the other, thus enabling the motorman of either locomotive to control the operation of the other.

Coupler Socket Lid. (Electric Locomotive.) 10, Figs. 4669-4670. A metal flap hinged at the top, to cover the outer end of a coupler socket.

Coupler Support. 2a, Figs. 3826-3829. A piece of cast or wrought iron fastened to a bumper and surrounding the shank or drawbar of a coupler to support and hold it in place. A **Drawbar Carry Iron,** which see.

Coupler Yoke. 3, Figs. 3826-3829. A pocket strap, U-shaped, which contains the spring and follower plates of a drawbar. It is the means of attaching the drawbar to the spring and follower plates, which are carried in the draft gear cheek plates riveted to the tender sills.

Coupling. "That which couples or connects, as a hook, chain or bar."—Webster. A coupling link was called simply a coupling. See **Coupler.**

Coupling Case. See **Hose Coupling Case.**

Coupling Hose. See **Air Brake Hose.**

Coupling Link. A wrought iron link or open bar by which engines and cars are coupled together by coupling pins. Chain coupling links are used with draw hooks. In consequence of the danger to trainmen attending the use of coupling links, and legislation forbidding their use after January 1, 1898, automatic couplers have almost entirely replaced them. See **Coupler.**

2. (British.) A link forming part of a coupling or draw chain. The open-ended link connected to the draw hook or drawbar is the coupling shackle. The intermediate links are sometimes termed the short links, and the end link the long link. A single long link is often used instead of three short intermediate links.

Coupling Nut. (Pipe Fittings.) A nut used to connect two pieces of pipe by what is called a union joint or **Union,** which see.

Coupling Nut Ring. A sleeve or collar forming part of a union joint. See **Union.**

Coupling Pin. A round bar of iron with which a coupling link is connected to a drawbar.

Coupling Pin Chain. A small chain attached to the car by a suitable eye to prevent the coupling pin from being lost.

Coupling Rod (British). 162, Figs. 5009-5010. See **Side Rod.**

Coupling Shackle (British). The end link of the coupling which is secured by a pin to the shank of the **Draw Hook,** which see.

Coupling Screw (British). A right and left-handed screw used in a **Screw Coupling,** which see.

Cover Plate. 1. A face plate of a steel-tired wheel connecting the tire and hub.

2. In metal underframes for tenders a plate which is riveted to the flanges of the center sills to give them additional vertical strength as a box girder. The plate riveted to the top flanges is called a top cover plate and one riveted to the bottom flanges a bottom cover plate.

Covering Plate, English for **Roof Sheet,** which see.

Cradle. See **Compressor Cradle.**

Crane. A machine for moving weights in two directions, vertically and horizontally. Figs. 5126-5129.

Crank. 1. "Literally a bend or turn; hence an iron axis with a part bent like an elbow, for producing a rotary motion or the contrary."—Webster. See **Bell Crank.**

2. An arm generally made in one piece with a driving wheel and having attached to it a pivot or pin called a crank pin or wrist pin to which a main or side rod is attached. In inside connected locomotives, including certain types of three-cylinder and four-cylinder compounds, the inside cranks are either forged in one piece with the axle, or preferably made separate and shrunk and keyed on. This latter form of construction is called a built-up crank axle.

Crank Axle. See above.

Crank Disk. 9, Figs. 1368-1385. One of two flat steel forgings, circular or of other shape, attached at right angles to a built-up driving axle, and having secured between them a crank pin. Also called crank web.

Crank Pin. Figs. 1130-1154. A short cylindrical shaft secured to a crank on a driving wheel, or joining the two webs or disks of a crank axle. The brass of a main rod or side rod is fitted to it, and its function is to transmit the reciprocating motion imparted to the piston, piston rod, crosshead and main rod by the expansive force of the steam, to the rotating driving wheel. Crank pins are distinguished as main crank pins if on the pair of driving wheels turned directly by the main; front crank pins if on the forward coupled or driving wheels; back crank pins if on the rear driving wheels, and front intermediate or back intermediate crank pins if on front or back intermediate driving wheels between the main driving wheels and the front or back drivers, as the case may be.

Crank Pin Bolt or Stud. 5, Figs. 1135-1139. A stud bolt set in the end of a crank pin and on which a nut can be screwed to hold a washer in place. See **Crank Pin Nut.**

Crank Pin Bushing. 2, Figs. 1155-1229. A sleeve or collar made of brass or anti-friction metal, forced into a suitable hole in the stub end of a main or side rod and forming a bearing for the crank pin.

Crank Pin Collar. 4, Figs. 1135-1139. A circular enlargement or fillet on a crank pin, either on that portion between two rod bearings, or between a rod bearing and the face of the crank.

Crank Pin Nut. 6, Figs. 1135-1139. A nut screwed on the outer end of a crank pin to secure a cap or washer that holds the rod in place on the pin.

Crank Pin Press. (Portable.) Fig. 5124. A portable hydraulic press for forcing crank pins in and out of locomotive driving wheels.

Crank Pin Truing Machine. (Automatic.) Fig. 5122.

Crank Pin Turning Machine. (Portable.) Fig. 5123. A machine for truing worn crank pins while in place on the driving wheel.

Crank Pin Washer or Collar. 4, Figs. 1135-1139. A circular plate with a hole in the center, held in place by a nut on the outer end of a crank pin to hold a stub end brass in place.

Crank Pin Wheel Seat. 1, Figs. 1135-1139. That portion of a crank pin that is in the body of a crank or wheel.

Crank Web. See **Crank Disk.**

Creco Brake Beam. Figs. 4456, 4452A.

Creco Journal Box. Fig. 4489.

Creco Side Bearings. Figs. 4517-4518.

Cross Bar (Swing Link Hanger). The iron bar sup-

pocket that encloses the draft spring and is bolted to the end of the drawbar is called a yoke.

Draw Gear. Figs. 3789-3819. The arrangement by which the engine transmits its power to pull the tender. A rod or bar, called a drawbar, is secured in the engine drawhead casting by a pin; and its other end rests in a similar casting secured to the forward end of the tender frame. See **Draft Connection** and **Draft Gear.**

Drawhead. Figs. 3817-3819; 2302-2409. A socket or receptacle for the end of the drawbar, which is held in place by a pin. The front engine drawhead is secured to the bumper, while the back engine drawhead, holding the tender drawbar, is usually cast in one piece with the deck or foot plate. Called in Great Britain a drag plate casting.

Draw Pin. Fig. 3796. A rod or pin passing through a drawhead and a drawbar to hold them together.

Draw Pin Key. A pin or cotter passing through a slot in the bottom end of the draw pin.

Draw Spring. See **Draft Spring.**

Drilling. A term used for **Switching,** which see, or making up trains. Regulating is another term sometimes used. The British term is marshalling or shunting.

Drilling Machine. Figs. 5132-5145. A machine for drilling holes. The different types are single, double, and multiple spindle drilling machines, radial drilling machines and semi-radial drilling machines.

Drip Cup (Air Brake). A receptacle inserted in the brake pipe of each car to receive water condensing therein. A drain cup.

Drip Valve. A globe or angle valve or a plug cock used to draw off the water of condensation from a pipe, pocket, cylinder or reservoir to prevent damage by freezing.

Driver. See **Driving Wheel.**

Driver Brake. Figs. 2590-2591. A brake, actuated by compressed air or vacuum apparatus, applying brake shoes to the driving wheels of a locomotive. They are either equalized brakes or **Cam Brakes,** which see. See **Air Brake.**

Driver Brake Bell Crank. A bell crank attached to a locomotive frame for transmitting the movement of the driver brake cylinder piston rod to the driver brake rod. It is pivoted on a heavy casting bolted to the frame and called a brake bell crank bracket.

Driver Brake Bell Crank Bracket. See **Driver Brake Bell Crank.**

Driver Brake Connecting Rod. 7, Figs. 2592-2594; 6, 7, 8, 9, Figs. 4664-4667. A steel rod fastened to the driver brake cylinder lever or a brake equalizer at one end, and to the brake hanger of a driving wheel at the other end. They are distinguished as front, second, third and fourth, according to the driving wheel to which they extend.

Driver Brake Connecting Rod Equalizer. A short metal bar to which the brake rod of one of the driving wheels is attached at a point between the ends. One end of the equalizer is secured by a short link to the hanger or lever of the forward driver brake, and the other end of the equalizer is secured to the second driver brake rod. In Figs. 2650-2652, the floating lever 9 acts as an equalizer.

Driver Brake Cylinder (Air Brake). 1, Figs. 2592-2594; Fig. 2497. The brake cylinder used to operate the driving wheel brakes. See **Brake Cylinder.**

Driver Brake Cylinder Angle. An angle casting bolted to the frame of a locomotive to support a driver brake cylinder. In some designs it is located at the back end of the frames, and in others is secured to the guide yoke or other frame crosstie.

Drive Spring Pocket (Electric Locomotive). See **Driving Pins.**

Driving Axle. 72, Figs. 80-124; Figs. 1368-1370. An **Axle,** which see, on which two coupled driving wheels are mounted. They are classified as front, intermediate, main and back. In four-cylinder compound locomotives and inside-connected engines, the driving axles become crank axles. Figs. 1371-1385.

Driving Axlebox (British). 76, Figs. 5009-5010. See **Driving Box.**

Driving Axlebox Spongebox (British). See **Driving Box Cellar.**

Driving Box. 70, Figs. 80-124; Figs. 1386-1445. A box-shaped casting of iron or steel, fitted to slide vertically in the jaws or pedestals of the frame, and holding a brass or bearing resting upon the driving axle journal. British, driving axlebox. See **Journal Box.**

Driving Box Bearing. 3, Figs. 1386-1394. The brass in a driving box. It rests on the journal of the driving axle.

Driving Box Cellar. Figs. 1401-1402; 73, Figs. 80-124. A receptacle in the bottom of a driving box, to hold oil and waste, secured to the box by two bolts called cellar bolts. British, driving axlebox spongebox.

Driving Box Cellar Bolt. 6, Figs. 1386-1394. A bolt passing through the bottom of the driving box longitudinally and through the oil cellar to hold it in place. Two are used for each box. British, driving axlebox spongebox bolt.

Driving Box Saddle. 151, Figs. 80-124; Figs. 1743-1751. A U-shaped piece or iron resting upon the driving box and forming a base for the support of a driving spring; or hanging downward to hold the spring below the box on engines having underhung springs. Also called driving spring stirrup and driving spring seat. British, driving axlebox saddle.

Driving Box Shoe. See **Driving Box Wedge.**

Driving Box Wedge. 219, Figs. 80-124; Figs. 1398-1400. A strip of iron or steel fitted to the inside of the legs or pedestals of the frame to protect them from wear by the sliding up and down of the driving boxes. The frame legs are usually built with one slightly inclined toward the other, and the wedges are made, one with parallel sides, and the other wedge-shaped. The wedge being against the sloping leg, can be moved vertically by set screws to take up the wear of the box. The wedge with parallel sides is called a shoe. British, driving axlebox wedge or shoe.

Driving Crank. See **Crank.**

Driving Horn Block (British). 74, Figs. 5009-5010. See **Pedestal.**

Driving Pins. (Electric Locomotive.) Figs. 4693-4696. Short steel pins resembling crank pins fastened on the end of the armature quill of the Westinghouse A. C.-D. C. electric locomotive. These pins enter cylindrical pockets formed in the inside face of the driving wheel hub, and transmit the motion of the armature to the wheel through the medium of springs inserted between pin and pocket.

Driving Spring. 80, Figs. 80-124; 1, Figs. 1605-1728; Figs. 1757-1780. A half-elliptic spring built up of flat steel plates and having a saddle or stirrup rest-

simple form of ball valve. An **Ejector**, which see, on the locomotive, maintains a vacuum of from 20 to 24 inches in the train pipe and in the cylinders under each car. In the release position the piston rests by its own weight in the bottom of the cylinder. To apply the brakes air is admitted to the train pipe and through the ball valve under each car to the space **below** the piston. The vacuum **above** the piston permits the atmospheric pressure below the piston to raise it and apply the brakes. A vacuum is always maintained above the piston and is available for applying the brakes at any time. In case the train parts the admittance of air to the broken train pipe applies the brakes in both sections of the train. A valve in the guards van may also be used to admit air to the train pipe and apply the brakes in case of emergency. To release the brakes the vacuum is restored in the train pipe and under the pistons by working the ejector.

Vacuum Valve. Figs. 1009-1011. A poppet valve inserted in the steam chest or cylinder head of a locomotive to relieve the vacuum formed when the engine is drifting—i. e., running with the throttle valve closed.

Valve. "A lid, plug or cover, applied to an aperture so that by its movement, as by swinging, lifting and falling, sliding, turning or the like, it will open or close the aperture to permit or prevent passage, as of a fluid."—Webster.

Piston Valve **By-Pass Valve** **Throttle Valve**
Slide Valve **Distributing Valve** **Intercepting Valve**
Vacuum Valve **Triple Valve** **Engineer's Brake Valve.**
Relief Valve **Reducing Valve**

Valve Body. The shell, case or frame of a valve.

Valve Follower. D, Figs. 2167-2192. A circular cast iron or steel end piece fastened to the body of a piston valve.

Valve Gear. Figs. 1807-2096. The mechanism by means of which the slide valves or piston valves are operated for the purpose of admitting steam to and exhausting it from locomotive cylinders. Valve gears are so arranged as to allow steam to be cut off at any desired point of the stroke of the piston, and admit steam to either end of the cylinders in order to run the engine either forward or backward. The ordinary link motion or so-called Stephenson link motion uses two eccentrics for each valve, the eccentric rods imparting motion to a link having a sliding block to which one end of a rocker arm is connected. The other arm of this rocker is connected to and thus moves the valve rod and valve. The point of cut off as well as the direction in which the engine turns its wheels is determined by the position of the link block and rocker pin in the link. The link is suspended by a link lifter attached to an arm on the reverse shaft. When this shaft is turned by the operation of the reverse lever, reach rod and reverse shaft arms, the link is raised or lowered, thereby moving the link block and rocker pin either backward or forward and consequently causing the valve to move on its seat. Inside connected engines generally have the block directly connected to the valve rod without the interposition of any rocker.

The Walschaerts valve gear, Fig. 1812, differs from the ordinary link motion in having only one eccentric rod, in altering the position or travel of the valve by moving the end of a bar attached to the end of the valve rod up or down in the link, without moving the link, and in having the valve rod also attached to a lever that derives its motion from the crosshead. This last detail determines the **Lap** and the **Lead**, which see, of the valve, and gives them a fixed value; whereas the ordinary link motion gives a variable lap and lead, affected by the valve travel. The Walschaerts gear derives its motion from an eccentric crank or return crank on the main crank pin, or from one eccentric on the main axle. The eccentric rod is secured to one end of the link which is pivoted in the center on a pin held by a bracket bolted to the guide yoke. The link block is secured to a radius arm or bar, one end of which is attached to the end of the valve rod and the other end to the lifting arm of the reverse shaft. The motion imparted to the valve by the crosshead connection is small, as the crosshead arm and union link are attached to the lower end of the combining or combination lever while the radius arm and valve rod are connected close to the upper end, thus imparting only a slight motion to the valve rod. For large locomotives, Walschaerts gear is now generally used, because it has lighter moving parts, and these parts are more accessible for inspection or repair than those of the common form of link motion. In the Gooch valve motion, the link is hung with its curvature reversed to its usual position, and in some cases is driven by eccentrics on an axle ahead of it.

Valve Handle. A hand grip attached to the plug, stem or movable portion of a valve for the purpose of opening and closing it.

Valve Packing Ring. 40, Figs. 2140-2164. One of four or more cast iron or composition rings secured on the outside of a piston valve to cause it to fit steam tight in the bushing or valve chamber. They are usually made slightly larger than the internal diameter of the valve chamber bushing, cut, and sprung in place, in the same manner as the packing rings on a steam cylinder piston. See **Piston**, and **Piston Valve.**

Valve Plate. See **Balance Plate.**

Valve Rod. Figs. 2083-2094; 36, Figs. 1813-1815. A steel shaft attached to a rocker arm or link block at one end, and to a valve stem at the other, for communicating the motion of an eccentric rod to a valve. British, slide valve intermediate spindle.

Valve Rod Block. 44, Figs. 2069-2075. A rectangular piece of metal attached to a rocker pin, and fitted to slide vertically in a rectangular frame or yoke formed on the valve rod, thus allowing the rocker pin to travel in the arc of a circle while the valve rod moves in a straight line parallel to its own axis. Also called Scotch yoke block.

Valve Rod Block Yoke. 198, Figs. 80-124; 43, Figs. 2069-2075. See above.

Valve Rod Guide. A cast iron support commonly bolted to the guide yoke, and having a hole through it in line with the valve rod stuffing box to keep the rod in its proper place.

Valve Rod Guide Bushing. A cylindrical lining or sleeve fitted in a valve rod guide to form a wearing surface for the rod.

Valve Rod Key. 37, Figs. 2069-2075. A wedge-shaped piece of metal inserted in an enlargemtnt on the inner end of a valve rod to secure it to the valve stem.

Valve Rod Knuckle. Figs. 2089-2094. A flexible joint connecting a valve stem to a valve rod. Since the rocker pin travels over an arc of a circle, and the valve stem in a straight line, some form of flexible connection must be used to permit each to move in its own path.

and holding a bearing or brass resting upon the journal of the truck axle. See **Journal Box.**

Engine Truck Box Cellar. 1, Figs. 3549-3596. Figs. 3587-3589. A receptacle in the bottom of an engine truck box, to hold oil and waste.

Engine Truck Box Cellar Bolt. 6, Figs. 3549-3596. A bolt passing through a truck box cellar to hold it in place.

Engine Truck Brake (Air Brake). The arrangement of brake cylinder, levers, beams and shoes for retarding or breaking the front engine truck wheels. Usually entirely separate from the driver brakes. Some locomotives have no brakes on the front truck.

Engine Truck Brake Cylinder (Air Brake). Fig. 2498; 2, Figs. 2592-2594. The brake cylinder used in connection with the air brake equipment on the forward truck of a locomotive. See **Brake Cylinder.**

Engine Truck Center Casting. 20, Figs. 3344-3351; Figs. 3352-3356. A casting bolted or riveted to the top of an engine truck and receiving directly the weight of the forward part of the engine. Also called a **Center Plate,** which see.

Engine Truck Center Pin. See **Center Pin.**

Engine Truck Center Pin Washer. 6, Figs. 3334-3351. A washer placed on the center pin to form a bearing between the head of the pin and the center plate. See **Center Pin.**

Engine Truck Equalizing Beam. British, bogie compensating beam. See **Equalizer.**

Engine Truck Frame. See **Frame.**

Engine Truck King Bolt. See **King Bolt** and **Center Pin.**

Engine Truck Oil Cup. A receptacle for holding oil and supplying it to the bearings of the engine truck boxes.

Engine Truck Pedestal. See **Pedestal** and **Truck Pedestal.**

Engine Truck Safety Chain. See **Safety Chain.**

Engine Truck Safety Chain Eye Bolt. An eye bolt in the front end of the frame on each side of the engine, for the attachment of a chain fastened to the forward corner of an engine truck to restrain its turning in case of derailment.

Engine Truck Side Bearing. 20, Figs. 3507-3525. A metal plate, block or set of rollers attached to the upper part of a four-wheel or six-wheel truck of a tank locomotive of the Forney type. It bears against a corresponding side bearing on the locomotive frame.

Engine Truck Side Bearing Pin. 21, Figs. 3507-3525. A roller forming part of an engine truck side bearing.

Engine Truck Spring. See **Truck Spring.**

Engine Truck Spring Seat. The place at which the weight of the forward end of a locomotive is transferred to the truck axles. At this place a spring band on each side of the engine truck is secured to the side frame.

Engine Truck Swing Bolster. A bolster from which the center plate of the truck is swung or suspended by means of short links hung on pivots or pins. It enables the truck to oscillate transversely to the center line of the engine, and thereby more readily adapt itself to the track when running on uneven or curved track. See Figs. 3376-3380.

Engine Truck Swing Frame. See **Engine Truck Swing Bolster.**

Engine Truck Wheel. See **Truck Wheel** and **Wheel.**

Engineer's Brake Valve (Air Brake). 235, Figs. 80-124; Figs. 2471-2482. A valve in the cab for operating the air brake. It has connections to the main reservoir, brake pipe and equalizing reservoir, as well as to the air gage and the atmosphere. Commonly called brake valve.

Engineer's Brake Valve Bracket. A support in the cab to hold the engineer's brake valve.

Engineer's Brake Valve Handle. 8, Figs. 2471-2473. The lever by which the engineer's brake valve is moved to its various positions.

Engineer's Brake Valve Spindle. 12, Figs. 2471-2473. The shaft moved by the handle and carrying the engineer's brake valve.

Equalizer. Figs. 1659-1677. A beam connected at each end to a driving or truck spring, or to the end of another similar beam, for the purpose of distributing the weight of an engine or tender to two or more axles, and to prevent an excessive load being thrown upon one axle by reason of inequalities of the track or roadbed. Called also equalizing beam. Locomotives having two driving axles have these two equalized together, while those with three or more commonly have the forward driving axle equalized with the leading truck. Equalizers are always used for four-wheel engine trucks, and frequently with tender trucks. British, compensating beam. Equalizers are designated as traverse equalizers when they connect the equalizing systems on the two sides of the locomotive. They are also designated by their position as equalizer, top of box; equalizer between drivers, etc.

Equalizer Fulcrum. 1. The point at which the equalizer or equalizing beam is supported.

2. A casting bolted to an engine frame to hold an equalizer at a point between its ends. 13, Figs. 1649-1650.

Equalizer Fulcrum Gib. A small wedge or strip passed through a slot in an equalizer fulcrum to hold the beam in place.

Equalizer Fulcrum Pin. A pin or bolt for holding an equalizer. It serves the same purpose as a gib.

Equalizer Fulcrum Plate. A casting or support, bolted on or between the frames, to hold an equalizer fulcrum.

Equalizer Hanger (Engine Truck). A link or support to connect the end of a truck spring with a point on the equalizer, usually near one end.

Equalizer Nut (Engine Truck). A nut on the end of the pin or bolt that passes through the equalizer hanger and the two equalizers on each side of the truck.

Equalizer Spring. A spring which rests on an equalizing bar and carries the weight of a tender. Single or double coil spiral springs are generally used for this purpose.

Equalizing Spring Cup. A casting on top of the spring, which bears against the under side of the wheel piece and holds the spring in its place.

Equalizer Spring Seat. A casting which sets on an equalizing bar and on which the spring rests. See **Spring Plate.**

Equalizer Strap. An iron band or holder bolted to the frame, through which the equalizer passes.

Equalizer Support. The hangers or suspenders that connect the ends of a transverse equalizer to the driving springs.

Equalizer Support Bracket. A casting bolted to a frame crosstie, to hold the fulcrum of the forward transverse equalizer in ten-wheel (4-6-0) locomotives where the drivers are not equalized with the forward truck.

Axle.	No. Blows.	Height of Drop.	Deflection.
M. C. B. 4¼ by 8 inch journals for 60,000-pound cars.....	5	34 feet	7 inches
M. C. B. 5 by 9 inch journals for 80,000-pound cars.....	5	48 "	5¾ "
M. C. B. 5½ by 10 inch journals for 100,000-pound cars..	7	48 "	4 "

8. Axles will be considered as having failed on physical test and will be rejected if they rupture or fracture in any way, or if the deflection resulting from the first blow exceeds the following:

M. C. B. axle, 4¼ by 8 inch journals..7½ inches
M. C. B. axle, 5 by 9 inch journals..6¼ inches
M. C. B. axle, 5½ by 10 inch journals..4½ inches

9. Axles will be considered to have failed on chemical test and will be rejected if the analysis of the borings taken as above described gives figures for the various constituents below, outside the following limits, namely:

Carbon..below 0.35 per cent., or above 0.50 per cent.
Manganese .....above 0.60 per cent.
Phosphorus .....above 0.07 per cent.

In order to measure the deflection, prepare a straight-edge as long as the axle, by reinforcing it on one side, equally at each end, so that when it is laid on the axle, the reinforced parts will rest on the collars of the axle, and the balance of the straight-edge not touch the axle at any place. Next place the axle in position for test, lay the straight-edge on it and measure the distance from the straight edge to the axle at the middle point of the latter. Then, after the first blow, place the straight-edge on the now bent axle in the same manner as before, and measure the distance from it to that side of the axle next to the straight-edge at the point farthest away from the latter. The difference in the two measurements is the deflection.

Axle Box (British). 46, Figs. 5009-5010. A **Journal Box,** which see. See also **Grease Axle Box, Oil Axle Box.**

Axle Box Bearing (British). See **Journal Bearing.**

Axle Box Cover (British). A hinged movable cover on the tender axle box through which the lubricant is introduced. On British oil axle boxes the cover is generally bolted to the box with a strip of leather interposed to make an oil-tight joint. The oil is replenished through a small orifice closed by a screw plug or spring hinge.

Axle Box Keep (British). The lower part of a tender axle box, which in an oil box, contains the lubricant, and in a grease box simply protects the under side of the journal from dust. See **Cellar.**

Axle Box Sponge Box (British). See **Driving Box Cellar.**

Axle Box Lid (British). See **Journal Box Lid.**

Axle Collar. 10, Fig. 3545. A rim or enlargement on the end of an axle which takes the end thrust of the journal bearing.

Axle Guard (British). American equivalent, pedestal. The ordinary or W pattern consists of a wrought iron plate attached to the sole bar, which permits vertical motion of the tender axle box, but restrains movement in any other direction.

Axle Guard Crown (British). The main part of the **Axle Guard,** which see.

Axle Guard Crown Washer (British). A piece of wrought iron plate, used as a washer for three or more bolts, which secure the main part of the axle guard to the tender sole bar.

Axle Guard Keep or **Horn Stay** (British). A piece of iron which secures the lower end of the jaws of the tender axle guards together.

Axle Guard Stay Rod or **Axle Guard Stretcher** (British). American equivalent, pedestal tie bar. A longitudinal bar connecting the lower ends of the tender axle guards, and keeping them the right distance apart.

Axle Guard Wing (British). The inclined part of an axle guard, strengthening it fore and aft.

Axle Guard Wing Washer (British). A piece of iron plate used as a washer for two or more bolts securing the wing of the axle guard to the sole bar. See **Axle Guard Wing.**

Axle Packing. A **Dust Guard,** which see. The journal packing is often called axle packing.

Axle Seat. 4, Figs. 3526-3540. The inside surface of the hole in a wheel which comes in contact with the axle, and not the hole itself. The corresponding part of an axle is called the wheel seat or wheel fit.

B

Babbitt Metal. "An alloy, consisting of 9 parts of tin and 1 of copper used for journal boxes; so called from its inventor, Isaac Babbitt, of Boston. Some variations have been made, and among the published compositions are:

Copper	1	1
Antimony	1	5
Tin	10	50

Another formula substitutes zinc for antimony. The term is commonly applied to any white alloy for bearings, as distinguished from the box metals or brasses in which copper predominates."—Knight.

Babbitt Metal Bearings. A style of bearing of which a great variety of forms exist, which in effect substitutes Babbitt metal in some of its many forms for brass as a bearing surface. **Lead-Lined Bearings,** which see, are different in that they merely use a thin sheet of lead over the brass, to correct slight irregularities and give an even bearing surface. The bearing or brass should be bored out to remove scale.

Back Bumper. 122, Figs. 79-124; Figs. 1531-1537. A cast iron or steel beam fastened across the frames under the deck at the back end of an engine. Also called tail piece, foot plate and deck plate.

Back Course (Stayless Boiler). 71, Figs. 151-152. The plate on a **Lentz** or **Vanderbilt Boiler,** which see, that corresponds to the outside firebox sheet of an ordinary locomotive boiler.

Back Cylinder Head. 1. (Air Brakes.) See **Non-Pressure Head.**

2. (Engine Cylinder.) Figs. 918-919, 922-925, 928-933. The **Cylinder Head,** which see, through which the piston rod passes; the head nearest the crosshead.

Back Face Plate (Steel Tired Wheels). The inner one of the two plates connecting the tire with the hub.

Back Head (Boiler). 129, Figs. 79-124; 4, Figs. 151-204, The plate forming the back end of a boiler. It is separated from the back firebox sheet by the back water space. See **Boiler.**

Back Head Braces. 9, Figs. 151-204; Figs. 265-269. Supports for the back head extending from the firebox side sheets. They are attached with pins to angle or channel brackets riveted to the inside face of the back head.

Back Head Brace Angle. 5, Figs. 151-204. A steel plate bent to a right angle and riveted inside the boiler back head to hold one end of a brace binding the back head to the firebox side sheet.

hand wheel on a shaft having a worm gear which moves an arm attached to the reach rod.

Reverse Lever Foot Rest. Figs. 1973-1975. An inclined step or brace attached to the reverse lever quadrant or other convenient place, against which the engineman can place his foot when moving the reverse lever.

Reverse Lever Fulcrum. Figs. 1999-2000. A casting having lugs or jaws and holding the lower end of a reverse lever by a pivot. The fulcrum is secured either to the deck plate or to the frame.

Reverse Lever Fulcrum Pin. Figs. 1997-1998. The pivot passing through the lower end of a reverse lever and holding it in place in the fulcrum.

Reverse Lever Handle. The upper end of a reverse lever, which is grasped by an engineman.

Reverse Lever Latch. 6, Figs. 1917-1951. A catch or detent having a tooth or bolt held in place by a latch box through which it passes, and forced into a notch in the reverse lever quadrant by means of a spring.

Reverse Lever Latch Box. A box-shaped piece fastened to the reverse lever to hold the latch bolt, which passes through it.

Reverse Lever Latch Gib. A steel wedge or strip in a reverse leyer latch box, held in place by a set screw or thumb screw, to form a guide for the latch bolt.

Reverse Lever Latch Link. 7, Figs. 1917-1958. One of two small rods fastened to the reverse lever latch at one end and to the latch handle at the other for the purpose of lifting the latch bolt out of a notch in the quadrant when the latch handle is grasped and pressed up against the reverse lever handle.

Reverse Lever Quadrant. 1, Figs. 1917-1958. A steel bar or bars forming an arc of a circle, over or along which the reverse lever moves. It is secured to the frame or deck plate or to the side of the boiler, and has notches cut in its upper surface for the bolt of the reverse lever latch.

Reverse Lever Quadrant Bracket. Figs. 1966-1971. A lug secured to the side of the boiler to serve as a bracket to which the quadrant is bolted.

Reverse Lever Quadrant Stand. A standard secured to the deck plate and having a lug or projection on its upper end to which the quadrant is bolted.

Reverse Lever Slot Plate. A piece of metal having a narrow longitudinal opening in it fastened to the floor of a locomotive cab to allow the reverse lever to pass through. A slot plate is used in designs where the reverse lever fulcrum is attached to the frame and is below the cab floor.

Reverse Shaft. 12, Figs. 2001-2006. A transverse shaft resting at each end in bearings or boxes bolted to the frames. The upper reverse arm, secured to one end of this shaft, rotates it when the reverse lever and reach rod move. The reverse shaft carries also two arms, one near each end, from which the link lifters or hangers are suspended. Called also lift shaft and lifting shaft. British, weigh shaft.

Reverse Shaft Bearing. Figs. 2013-2017. A casting bolted on the frames on each side, with a cap bolted on, furnishing a bearing for the ends of the reverse shaft.

Reverse Shaft Bearing Bushing. A bushing or sleeve in the reverse shaft bearing, usually case hardened, and forming the actual bearing for the reverse shaft.

Reverse Shaft Bearing Cap. A cap or cover bolted on top of a reverse shaft bearing.

Reverse Shaft Box Bolt. A bolt holding the reverse shaft box or bearing in place on the frame.

Reverse Shaft Pin. A pivot or pin on the lower arm of a reverse shaft, and on which the upper end of the link lifter is secured.

Reverse Shaft Reach Rod. See **Reach Rod.**

Reverse Shaft Reach Rod Guide. A guide, strap or keeper through which the reach rod passes, to hold it in place. Usually secured to the running board or the side of the boiler.

Reverse Shaft Reach Rod Pin. A pin or pivot connecting the reach rod with the reverse shaft arm.

Reverse Shaft Spring. Figs. 2024-2025; 9, Figs. 2032-2037. A half elliptic or helical spring, attached to a lug or short arm on the reverse shaft. The tension of the spring balances the weight of the links and eccentric rods and assists the engineman in moving the reverse lever.

Reverse Shaft Spring Bracket. Figs. 2026-2031. A transverse iron brace bolted to the frames to hold the reverse shaft spring. Called also reverse spring bracket.

Reverse Shaft Spring Casing. 9a, Figs. 2032-2037. A cylindrical cover or housing for a coil or helical reverse spring.

Reverse Shaft Spring Lug. Figs. 2038-2039. A lug attached to the center of a reverse shaft for the attachment of the spring rod by means of a pin.

Reverse Shaft Spring Nut. A nut on the reverse spring rod to adjust the tension exerted by the spring on the reverse shaft.

Reverse Shaft Spring Pin. A bolt holding one end of the spring rod securely to the spring lug.

Reverse Shaft Spring Rod. 11a, Figs. 2032-2037. A rod fastened to a reverse shaft spring and connecting it with the lug on the shaft arm.

Reverse Spring. See **Reverse Shaft Spring.**

Reversing Cock (Air Brakes). Fig. 2534. A special form of cock used with the high-speed brake and double-pressure control equipments, for the purpose of throwing the control of brake pipe pressure on to either one of the two slide-valve feed valves attached to its sides. One feed valve is set for a low-brake pipe pressure, and the other for a high. The reversing cock is connected by piping to the brake valve in such a way that when feeding up the brake-pipe pressure the air must pass through the reversing cock and one of the feed valves before going to the brake pipe. The feed valve through which it passes depends on the position of the reversing-cock handle.

Reversing Cock Pipe Bracket (Air Brake). Fig. 2536. A bracket arranged for pipe connections, which bolts to the brake valve in place of the slide valve feed valve, and is connected by piping with the **Reversing Cock,** which see. Often called a feed-valve pipe bracket.

Reversing Valve (Air Pump). 72, Figs. 2429-2435. A slide valve operated by the reversing rod in a small cylinder in the top head, which controls the admission and exhaust of steam to the chamber back of the large main-valve piston, thereby causing the main valve mechanism to operate. See **Reversing Valve Rod.**

Reversing Valve Bush (Air Pump). 73, Figs. 2429-2435. The bushing which is inserted in the top head and forms the seat for the reversing valve. See **Bushing.**

Reversing Valve Chamber Cap (Air Pump). 74, Figs. 2429-2435. A screw plug by means of which access

cylindrical in form and projects beyond the disks or spokes of the wheel on each side. In Great Britain termed the boss.

Hydraulic Jack. Figs. 4599-4608. A tool or machine in which the power is exerted by means of the pressure of some liquid acting against a piston or plunger, for raising heavy weights, like an engine. The head and interior tube or ram form a reservoir, from which the fluid flows to the pump, and to which it is returned in lowering. From the pump it is forced, by the downward stroke of the piston, past the lower valve into the cylinder, and, this being closed at the bottom, the ram rises. The lever, which is made with a projection on one edge, slips into a socket at the side of the head. This socket passes through an arm on the interior of the head, and to this is fastened the piston of the pump. The claw attachment is a third tube, which screws into the head, below the ram collar and outside the cylinder, at the lower end of which is a claw projecting out at one side. They are rated so that one man can raise the weight for which they are designed. The speed of lifting is inversely proportional to the amount lifted. Ten tons can be lifted one foot in about a minute and a half.

I

I-Beam. A general term applied by makers to any form of rolled iron or steel having an I cross-section. The top and bottom parts are termed the flanges, and the middle the web. The usual dimensions are given by the total height from out to out, and vary from 3 to 24 in. When one of the flanges is simply a round bar it is termed a deck beam. I-beams are used for center and intermediate sills, also for body and truck bolsters on tenders.

Independent Brake Valve (Air Brake). Figs. 2474-2476. A rotary, straight-air brake valve used with the Westinghouse ET locomotive brake equipment for operating the locomotive brakes independently of the train brakes. See **Engineer's Brake Valve.**

India Rubber. A gum which exudes from a tropical tree growing in the East and West Indies, Mexico, South America, etc. It is prepared for use by vulcanizing with a greater or less proportion of sulphur according to the stiffness required.

Indicator. A device attached to the cylinder of a steam engine, which gives a graphical diagram of the steam distribution in the cylinder at all points in the stroke. Used chiefly for testing purposes.

Indicator Fitting. A brass fitting with a three-way cock and pipe connections for an indicator from each end of the locomotive cylinder.

Indicator Plug. Figs. 949-951. A plug screwed into a hole in the side of a cylinder near the end. When the indicator is to be used this plug is removed and the indicator pipe connection screwed in. A hole is drilled near each end of a cylinder so that an indicator diagram may be taken from either end at will.

Injector. Figs. 3088-3107. An instrument for forcing water into a steam boiler in which a jet of steam imparts its velocity to the water and thus forces it into the boiler against the boiler pressure. The injector was first reduced to practical form by Henri Giffard, an eminent French engineer, in the year 1858, and was introduced in this country by William Sellers & Co., Inc., in 1860. See **Inspirator, Metropolitan Injector, Nathan Injector, Simplex Injector, Sellers' Injector.**

Injector Bracket. Figs. 2999-3000. A projection secured to the boiler for the attachment of an injector.

Injector Check. Figs. 3108-3110. A check valve placed in an injector delivery pipe and opening against boiler pressure to admit water forced in by an injector.

Injector Check Valve. See **Injector Check.**

Injector Connecting Rods. On injectors located outside the cab, rods secured to the steam and water valves and passing through the front of the cab so that the injector may be operated from inside.

Injector Delivery Pipe. 50, Figs. 80-124. A copper or iron tube through which feed water is forced into a boiler. When the pipe is located outside the boiler, the check valve is put at the farther end of the pipe; but with injectors secured to the boiler back head, a check valve is placed close to the injector, and the delivery pipe carried inside the boiler to a point near the front end. Also called an injector check pipe.

Injector Delivery Pipe Hanger. 51, Figs. 80-124. A hanger or suspender fastened inside a boiler to support the check or delivery pipe of an injector. Used when the injectors are on the boiler backhead and deliver water through a pipe terminating near the front end.

Injector Feed Pipe. 185, Figs. 80-124. A tube secured to the water inlet of an injector and connected to the hose from the tender.

Injector Feed Pipe Coupling. A coupling or union on the injector feed pipe to connect it to the hose from the tender tank.

Injector Hole. An aperture either in the first course of a boiler near the smokebox, or in the back head, for the injector delivery pipe to enter the boiler.

Injector Steam Valve. A valve usually arranged to take steam from the cab turret, and having a pipe through which steam is conducted to an injector.

Inlet Valves (Air Pump). 86, Figs. 2429-2435. The disk or poppet valves through which air enters the pump.

Inside Admission Valve. Figs. 2156-2157. A valve which takes live steam through a central cavity and admits it to the steam ports leading to the ends of the cylinder. The exhaust ports are at the ends of the valve. See **Valve.**

Inside Connected Locomotive. A locomotive having the main rods connected to cranks formed or built on the driving axle between the frames. Some four-cylinder compounds have two cylinders inside connected. In Great Britain many, if not most, simple engines are inside connected.

Inside Lap (Valve). With an outside admission valve, such as an ordinary slide valve, the inside lap is the amount by which the inside edge of the valve overlaps or goes beyond the inner edge of either steam port when the valve is at the exact center of its travel on the valve face. The effect of inside lap is to delay the release of the steam from the cylinder to the exhaust pipe. Sometimes called exhaust lap. See **Outside Lap.**

Inside Stack (Smokeboxes). 5, Figs. 343-442. See **Smoke Stack Lift Pipe.**

Inside Throat Sheet. 29, Figs. 151-206. A piece or plate forming part of the front end of a firebox in boilers having the back tube sheet set forward of the front water leg, and being therefore shaped so as to correspond with the outside throat sheet. See **Throat Sheet.**

Inside Welt. 56, Figs. 151-206. A strip or narrow plate